

Module 4 Lab 2: Textbook Problem 3.1

CS 82A - Introduction to Data Science

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GitHub: https://github.com/ramishatasfia5156-commits/cs82a-coursework/blob/main/module4/module4_lab2.ipynb

Load the Data

```
In [1]: import pandas as pd

df = pd.read_csv('allbacks.csv', index_col='rownames')
print('Shape:', df.shape)
df
```

Shape: (15, 4)

```
Out[1]:
```

	volume	area	weight	cover
rownames				
1	885	382	800	hb
2	1016	468	950	hb
3	1125	387	1050	hb
4	239	371	350	hb
5	701	371	750	hb
6	641	367	600	hb
7	1228	396	1075	hb
8	412	0	250	pb
9	953	0	700	pb
10	929	0	650	pb
11	1492	0	975	pb
12	419	0	350	pb
13	1010	0	950	pb
14	595	0	425	pb
15	1034	0	725	pb

Weight is in grams, volume in cubic centimeters. `area` is 0 for every paperback in this file (it differs from the textbook table), but area is not used in this

problem.

Median Cover Type (Textbook Part A)

```
In [2]: # A median works on ordered categories: sort, then take the middle value
cover_order = pd.CategoricalDtype(['hb', 'pb'], ordered=True)
covers =
df['cover'].astype(cover_order).sort_values().reset_index(drop=True)

print(df['cover'].value_counts())
print()
middle = (len(covers) - 1) // 2          # 15 books -> position 8 of 15
(index 7)
print('Middle position:', middle + 1, 'of', len(covers))
print('Median cover:', covers[middle])
```

```
cover
pb     8
hb     7
Name: count, dtype: int64
```

```
Middle position: 8 of 15
Median cover: pb
```

Analysis: There were 7 hardback copies that come before the 8 paperback copies; therefore, the 8th book out of 15 would be one of the paperback copies. Therefore, the median cover type in this sample is paperback. If we reversed the order and put the paperbacks first, the answer would still be the same because paperbacks make up more than half of the books.

Mean Weight (Textbook Part B)

```
In [3]: print(f"Mean weight, all books: {df['weight'].mean():.2f} g")
print()
by_cover = df.groupby('cover').agg(books=('weight', 'size'),
                                   mean_weight_g=('weight', 'mean'),
                                   mean_volume_cm3=('volume',
                                                  'mean')).round(2)
by_cover
```

```
Mean weight, all books: 706.67 g
```

```
Out[3]:
```

	books	mean_weight_g	mean_volume_cm3
--	-------	---------------	-----------------

cover			
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hb	7	796.43	833.57
pb	8	628.12	855.50

```
In [4]: diff = by_cover.loc['hb', 'mean_weight_g'] - by_cover.loc['pb',  
        'mean_weight_g']  
        print(f'Hardbacks are {diff:.2f} g heavier on average ({diff /  
        by_cover.loc["pb", "mean_weight_g"]:.0%} heavier than paperbacks)')
```

Hardbacks are 168.31 g heavier on average (27% heavier than paperbacks)

All Books: We can estimate that the average weight for a book in this sample is approximately 707 grams.

By Cover Type: The average weights for hardback books and paperback books were respectively around 796 g and 628 g. Therefore, there is an approximate difference of 168 grams (27%) between the two cover types, with hardbacks being heavier. Although both have nearly equal average volumes (hardbacks = 834 cm³; paperbacks = 856 cm³), the additional weight of the hardbacks comes from the cover type rather than the hardbacks being larger books. The library may find paperbacks beneficial because their lighter weight would make handling them and stocking shelves easier. However, using only 7 hardbacks and 8 paperbacks as a sample is too limited to base a long-term policy on.

Variance in Book Volume (Textbook Part C)

```
In [5]: print(f"Variance of volume, all books: {df['volume'].var():.2f} (cm3)2")  
        print()  
        df.groupby('cover')['volume'].agg(['mean', 'var', 'std']).round(2)
```

Variance of volume, all books: 115,518.35 (cm³)²

```
Out[5]:
```

	mean	var	std
cover			
hb	833.57	114100.62	337.79
pb	855.50	132979.71	364.66

The standard deviation for book volume is approximately 340 cm³ overall. The spread is also similar for both cover types; therefore, neither hardbacks nor paperbacks have a narrower range of book sizes. The variability in book volume between the two cover types is very similar.

Recommendation

The library should select paperback. This is the median cover type in the current sample as well as the more common one. Additionally, even though paperbacks and hardbacks are nearly the same in volume, a paperback weighs about 168 g

less than a hardback. The range of book sizes is also about the same for both cover types.

By-Hand Check: Mean Weight of All Books

Weights: 800, 950, 1050, 350, 750, 600, 1075, 250, 700, 650, 975, 350, 950, 425, 725

Sum = 10,600 g

Mean = 10,600 / 15 = **706.67 g**

```
In [6]: weights = [800, 950, 1050, 350, 750, 600, 1075, 250, 700, 650, 975, 350, 950, 425, 725]
print('Sum: ', sum(weights))
print('Mean:', round(sum(weights) / len(weights), 2))
print('Matches pandas:', round(sum(weights) / len(weights), 2) == round(df['weight'].mean(), 2))
```

Sum: 10600

Mean: 706.67

Matches pandas: True

AI Disclosure

AI Disclosure: Claude (Anthropic) was used to help explain concepts and draft initial code for this lab. All code was reviewed, retyped, and understood by me. I can explain how a median works on ordered categories like cover type, how groupby compares the mean weight of hardbacks and paperbacks, what the variance in book volume shows, and how those results support the cover recommendation.

Key prompt: "Help me with Module 4 Lab 2: Textbook Problem 3.1 using allbacks.csv."